Semantic Voice Compression for LoRa Communication in Infrastructure-Less Environments

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***Abstract*—Infrastructure-less environments, such as disaster zones and remote communities, impose severe bandwidth and power constraints that prevent conventional systems from supporting human-scale voice communication. LoRa, a low-power wide-area radio technology, offers extended coverage but is limited to 11–242 bytes per packet at data rates as low as 293 bps, making raw audio transmission entirely impractical. We present a semantic voice compression architecture that encodes speech using EnCodec, a neural audio codec based on Residual Vector Quantization (RVQ), and transmits the resulting compact token payload over a peer-to-peer LoRa link. EnCodec operating at 48 kHz with two RVQ codebook layers produces 15,000 integer codes for a 10-second clip, totaling approximately 4 kB—a 99.6% reduction from the 946 kB raw PCM file. This cuts the required LoRa packet count from 7,391 to 32 and transmission time from over 100 minutes to 25 seconds at SF7/125 kHz, well within FCC Part 15.247 duty-cycle limits. The system was validated on two Heltec ESP32S3 LoRa V3 modules with a custom three-phase fragmentation and session protocol. Reconstructed speech remained intelligible with speaker identity preserved; observed artifacts were consistent with low-bitrate neural coding and did not impair comprehension. This work shows that RVQ-based neural audio compression enables voice communication over severely bandwidth-constrained LoRa links devoid of centralized infrastructure, while retaining prosody and urgency cues that text-only alternatives discard.**

***Index Terms*—LoRa, semantic communication, neural audio codec, EnCodec, residual vector quantization, infrastructure-less networks, voice compression, ESP32**

I. Introduction

Environments lacking communication infrastructure, whether due to geographic isolation, natural disaster, or systemic limitation, often fail to support what we define as human-scale interaction: the effective transfer of intent and urgency between parties [1]. Modern communication systems are optimized for high-bandwidth settings where raw audio can be transmitted with minimal constraints. In infrastructure-less settings, strict limits on bandwidth, power, and network topology prevent conventional systems from preserving the nuance and semantic clarity required for meaningful communication.

Also, in many parts of the world, infrastructure is still limited or nonexistent. Rural villages, disaster-affected regions, and mountainous communities often lack cellular connectivity, making traditional communication methods ineffective. Conventional voice systems, such as walkie-talkies, also incur substantial energy costs that directly affect battery lifetime [2]. Long-range, low-power technologies such as LoRa (Long Range) were specifically designed for low-power, wide-area IoT deployments [3], but their strict payload limits of 11–242 bytes per packet make raw audio transmission entirely impractical: a 10-second speech clip at 48 kHz requires over 7,000 packets and more than 100 minutes of airtime under typical LoRa settings.

Text-only alternatives, such as speech-to-text (STT) pipelines, reduce bandwidth but discard prosody, emotion, and urgency cues. These are precisely the qualities that matter most in emergency communication. Our approach instead uses a neural audio codec to compress the audio representation itself, preserving the voice while eliminating the redundancy that makes raw transmission infeasible [1][4].

To address this gap, we propose a low-power, intelligent communication system designed not only to transmit data efficiently but also to preserve the essential urgency and communicative richness of human interaction under severe constraints using semantic voice compression.

This research proposes a semantic-aware voice communication system for decentralized LoRa mesh edge networks. We use semantic compression to encode a message's intent and knowledge rather than transmitting dense raw audio. Specifically, we employ a dual-stream semantic encoder and a generative neural vocoder to reconstruct intelligible speech while preserving speaker characteristics, prosody, and urgency cues that are often lost in STT-TTS pipelines [4]. By distributing semantic encoders and decoders across mesh-capable edge nodes, the system reduces dependence on centralized infrastructure while remaining compatible with strict LoRa payload and duty-cycle constraints.

Using the semantic voice compression algorithm, we achieved a 99.6% compression rate, reducing our packet count from 7,391 to 32 in a low-bandwidth LoRa communication system. This is possible through a neural audio codec, EnCodec, which operates natively at the original audio sample rate (48 kHz) and encodes speech into a compact set of discrete token indices that can be transmitted over our peer-to-peer LoRa network, with the model pre-installed at both ends for local reconstruction.

The key contributions of this paper are:

- A semantic compression architecture combining a neural audio codec with a peer-to-peer LoRa transport protocol.
- A validated implementation on real hardware demonstrating 99.6% compression, reducing packet count from 7,391 to 32 for a 10-second, 48 kHz speech clip.

- A reliable fragmentation and session protocol for audio delivery over lossy, low-bandwidth radio links.

The remainder of this paper is organized as follows: Section II reviews related work, Section III describes the system architecture, Section IV presents implementation and validation, and Section V concludes the paper.

II. Related Work

A. LoRa Technology for Low-Power Wide-Area Networks

LoRa is a chirp spread spectrum modulation that enables long-range, low-power communication, originally designed for sensor-class IoT devices [3]. Its physical layer supports spreading factors (SF) from 7 to 12 and bandwidths of 125, 250, or 500 kHz. Higher spreading factors improve receiver sensitivity and extend range at the cost of lower data rate and longer time-on-air; SF7 at 125 kHz yields the highest throughput of approximately 5.47 kbps, while SF12 extends range to several kilometers at the cost of reducing throughput to 293 bps [3]. The 915 MHz ISM band (902–928 MHz) used in this work is governed by FCC Part 15.247, which imposes a 400 ms maximum transmission window and 1% duty-cycle limit per channel.

These constraints make LoRa well-suited for low-rate telemetry but fundamentally incompatible with conventional voice codecs. Standard audio at 48 kHz PCM requires roughly 768 kbps, six orders of magnitude above the LoRa data rate ceiling, requiring a compression approach that reduces not just the bitrate but the number of transmitted symbols to within the duty-cycle window.

B. Human-Scale Communication in Infrastructure-Free Environments

Arregui Almeida et al. [5] presented a gateway-free LoRa mesh network on ESP32, demonstrating design, self-healing mechanisms, and empirical performance. Their results show reliable multi-hop delivery without a centralized infrastructure on the ESP32-SX1262 platform.

Mari and Gabillon [6] introduced the MauMe network, a LoRa multi-hop collaborative protocol with a low-cost implementation example. Both works validate decentralized LoRa mesh feasibility but do not explore semantic compression or voice communication, which are central to our system.

C. Voice Over LoRa

Text-only systems are prone to high word error rates in noisy disaster zones and can place a heavy cognitive burden on users who must read and interpret instructions during a crisis. Sending text or using STT-TTS removes emotion, prosody, and urgency signals cues, leading to loss of communicative richness during constrained emergency conditions [1].

Kagai et al. [7] demonstrated voice transmission over LoRa using a novel audio codec, showing that voice-over-LoRa is physically feasible. However, their approach relies on denser audio payloads and does not address the packet-count and duty-cycle constraints imposed by the ISM band. Our work uses neural codec compression to reduce the payload to 32 packets for a 10-second clip, bringing transmission within practical LoRa limits.

Urazayev et al. [4] proposed a LoRa-based voice communication system using speech-to-text (STT) with optional semantic compression via a large language model, greatly reducing transmission overhead in comparison to raw audio. However, their approach discards prosody and speaker identity through the STT stage, and their evaluation does not examine transmission range or network topology. In contrast, our work preserves the acoustic waveform using neural codec compression and validates the approach on real hardware in a peer-to-peer LoRa link.

D. Neural Audio Codecs and Semantic Compression

Neural audio codec research was pioneered by SoundStream [8], which introduced a fully convolutional encoder-decoder architecture with Residual Vector Quantization (RVQ) trained end-to-end. SoundStream demonstrated that a single model can span variable bitrates from 3 kbps to 18 kbps by adjusting the number of active RVQ layers at inference time, setting a template that EnCodec and subsequent codecs follow [8]. EnCodec [9] extended this architecture to 48 kHz stereo audio and added a language model-based entropy coder, achieving higher perceptual quality at equivalent bitrates.

SemantiCodec features a dual-encoder architecture: a semantic encoder using a self-supervised pre-trained Audio Masked Autoencoder (AudioMAE), discretized using k-means clustering on extensive audio data, and an acoustic encoder to capture remaining detail [10]. This design encodes more semantic information with fewer token sequences.

Traditional deep-learning-based joint source-channel coding (JSCC) methods do not focus on semantic similarity between source and reconstruction and mostly rely on MSE or SSIM/MS-SSIM as end-to-end distortion measures [11].

AudioMAE provides conceptual grounding for semantic extraction: its high masking ratio forces learning of global audio structure, the audio essence, rather than local fluctuations, allowing systems to focus on latent tokens representing sender intent [12].

Our current EnCodec implementation operates at the acoustic level; the proposed dual-stream semantic encoder would extend this to explicitly capture intent and prosodic content.

III. System Architecture

The system targets infrastructure-less communication situations where bandwidth, power, and duty-cycle constraints prevent the use of conventional voice pipelines. The architecture combines AI-based compression and low-bandwidth transport as a two-stage pipeline: a semantic compression module encodes raw speech into a compact token payload, and a LoRa transport layer delivers that payload over a peer-to-peer radio link. Both ends share the same pre-installed neural codec model, enabling local reconstruction without cloud connectivity. Figure 1 shows the end-to-end architecture.

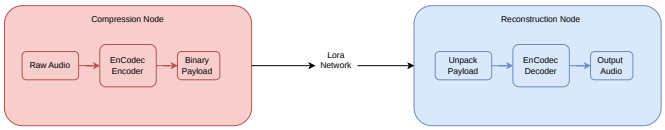


Fig. 1. Neural Codec Voice Compression System. The transmitting node encodes raw speech into discrete tokens via EnCodec. Tokens are packetized and sent over the LoRa P2P link. The receiving node reconstructs intelligible speech using the mirrored decoder.

A. Neural Codec Compression Module

The compression module maps raw waveform audio to a compact binary payload using EnCodec [9], a neural audio codec that operates natively at 48 kHz. EnCodec builds on the RVQ framework introduced by SoundStream [8] and extends it to higher sample rates with an additional language model entropy coder. The model has three stages: (1) a convolutional encoder compresses the waveform to a dense latent stream at 75 frames per second, (2) a Residual Vector Quantization (RVQ) module maps each frame to one or more integer codes from a 1024-entry codebook, and (3) a mirrored convolutional decoder reconstructs the waveform from those codes deterministically [9].

In RVQ, a cascade of n vector quantizers encodes the latent at each time step: the first quantizer selects the nearest codebook entry, and each subsequent quantizer encodes the residual error left by the previous one. This multi-level scheme allows variable bitrates by changing n at inference time without additional training. At $n = 2$ codebook layers (the configuration used in this work), the resulting token sequence encodes 4 kB for a 10-second clip—well within practical LoRa payload budgets. This module performs acoustic-level compression; full semantic encoding that separately captures linguistic intent and prosodic content is left for future work.

Before encoding, segment-level amplitude normalization stabilizes the encoding scale. The normalization scalar is transmitted in the packet header as a 2-byte float16 and used by the receiver to restore the correct amplitude. Because reconstruction depends only on the received token indices and this scalar, decoding is fully deterministic and avoids the phase-reconstruction artifacts introduced by spectrogram-inversion methods such as Griffin-Lim.

B. LoRa Transport Layer

The transport layer is a half-duplex peer-to-peer LoRa link operating at Spreading Factor 7 (SF7) and 125 kHz bandwidth in the 915 MHz ISM band [3]. SF7 was selected to improve throughput (≈5.47 kbps) and minimize time-on-air for the compressed payload; the trade-off of lower sensitivity is acceptable at the short indoor range of this validation. The 11–242-byte payload ceiling per packet requires fragmentation for any audio payload exceeding a single frame.

To deliver fragmented audio payloads reliably over a lossy link, the system uses a three-phase session protocol:

1) An `AudioStart` packet carries session metadata: fragment count, codec identifier, sample rate, audio duration, and a CRC16 checksum.
2) The receiver returns an ACK to authorize the transfer.
3) The transmitter sends sequential `AudioData` packets in fixed 255-byte frames (10-byte protocol header, up to 245 bytes of payload). An `AudioEnd` packet closes the session with a CRC32 checksum for end-to-end integrity verification.

ACK status codes distinguish success, CRC failure, and missing fragments. The `codec_id` field supports both raw PCM and EnCodec payloads, permitting direct comparison under identical link conditions.

IV. Implementation and Validation

A. Hardware

The system was implemented on two Heltec ESP32S3 WiFi LoRa V3 modules, each equipped with an SX1262 radio and a MicroSD card reader for storing transmitted files and capturing protocol logs. Figure 2 shows the breadboard hardware setup, and Figure 3 shows a transmission log confirming fragment delivery.

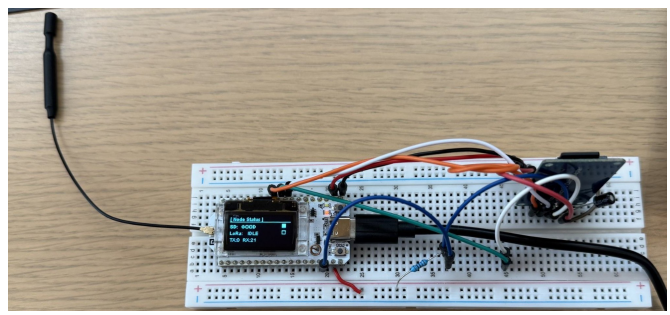


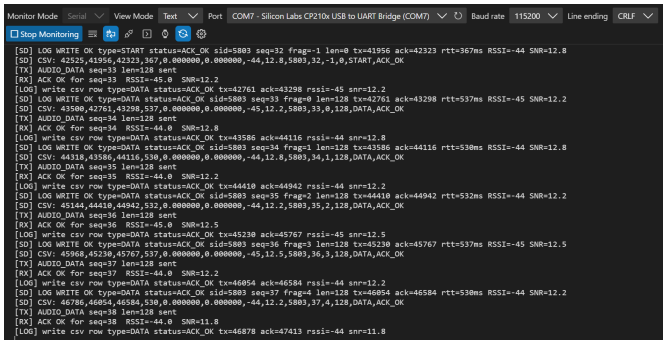
Fig. 2. Single-unit view of one of the two Heltec ESP32S3 LoRa V3 modules used in the peer-to-peer setup, with microSD interface wiring.

B. Software

For the development, we used VS Code with the Arduino extension to program the ESP32S3 modules. The LoRa communication protocol was implemented using the RadioLib library, which provides a high-level interface for the SX1262 radio. The fragmentation and session state machine runs entirely on the ESP32S3 without an operating system, using interrupt-driven reception and a finite-state machine for session control.

C. EnCodec Integration

EnCodec inference ran in Google Colab (NVIDIA T4 GPU) during this validation phase; the LoRa nodes handled all fragmentation, session management, and radio transport independently. On-device inference is a future milestone.



```
Monitor Mode  Serial   View Mode  Text   Port  COM7 - Silicon Labs CP210x USB to UART Bridge (COM7)   Baud rate  115200   Line ending  CRLF
Stop Monitoring
[SD] LOG WRITE OK type=START status=ACK_OK sid=5803 seq=32 frag=-1 len=0 tx=41956 ack=42323 rtt=367ms RSSI=-44 SNR=12.8
[SD] CSV: 42525,41956,42323,367,0.000000,0.000000,-44,12.8,5803,32,-1,0,START,ACK_OK
[TX] AUDIO_DATA seq=33 len=128 sent
[RX] ACK OK for seq=33  RSSI=-45.0  SNR=12.2
[LOG] write csv row type=DATA status=ACK_OK tx=42761 ack=43298 rssi=-45 snr=12.2
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=33 frag=0 len=128 tx=42761 ack=43298 rtt=537ms RSSI=-45 SNR=12.2
[SD] CSV: 43500,42761,43298,537,0.000000,0.000000,-45,12.2,5803,33,0,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=34 len=128 sent
[RX] ACK OK for seq=34  RSSI=-44.0  SNR=12.8
[LOG] write csv row type=DATA status=ACK_OK tx=43586 ack=44116 rssi=-44 snr=12.8
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=34 frag=1 len=128 tx=43586 ack=44116 rtt=530ms RSSI=-44 SNR=12.8
[SD] CSV: 44318,43586,44116,530,0.000000,0.000000,-44,12.8,5803,34,1,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=35 len=128 sent
[RX] ACK OK for seq=35  RSSI=-44.0  SNR=12.2
[LOG] write csv row type=DATA status=ACK_OK tx=44410 ack=44942 rssi=-44 snr=12.2
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=35 frag=2 len=128 tx=44410 ack=44942 rtt=532ms RSSI=-44 SNR=12.2
[SD] CSV: 45144,44410,44942,532,0.000000,0.000000,-44,12.2,5803,35,2,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=36 len=128 sent
[RX] ACK OK for seq=36  RSSI=-45.0  SNR=12.5
[LOG] write csv row type=DATA status=ACK_OK tx=45230 ack=45767 rssi=-45 snr=12.5
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=36 frag=3 len=128 tx=45230 ack=45767 rtt=537ms RSSI=-45 SNR=12.5
[SD] CSV: 45968,45230,45767,537,0.000000,0.000000,-45,12.5,5803,36,3,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=37 len=128 sent
[RX] ACK OK for seq=37  RSSI=-44.0  SNR=12.2
[LOG] write csv row type=DATA status=ACK_OK tx=46054 ack=46584 rssi=-44 snr=12.2
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=37 frag=4 len=128 tx=46054 ack=46584 rtt=530ms RSSI=-44 SNR=12.2
[SD] CSV: 46780,46054,46584,530,0.000000,0.000000,-44,12.2,5803,37,4,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=38 len=128 sent
[RX] ACK OK for seq=38  RSSI=-44.0  SNR=11.8
[LOG] write csv row type=DATA status=ACK_OK tx=46878 ack=47413 rssi=-44 snr=11.8
```

Fig. 3. Serial monitor transmission log showing successful fragment delivery for compressed payload transfer.

D. Experimental Setup

Tests were conducted using Spreading Factor 7 (SF7) and 125 kHz bandwidth in the 915 MHz ISM band, consistent with the LoRa parameter study by Augustin et al. [3]. SF7 maximizes throughput (≈5.47 kbps) and minimizes time-on-air for the compressed payload, while providing sufficient link budget for short-range indoor operation. Two conditions were compared under identical link settings:

- Raw audio: a 10-second speech clip at 48 kHz PCM (946 kB).
- Compressed audio: the same clip encoded with EnCodec at two RVQ codebook layers (4 kB token payload).

Both conditions used the same 255-byte frame structure and session protocol, differing only in payload content. Table I summarizes the LoRa radio configuration used in all tests.

TABLE I
LoRa Radio Configuration

Parameter	Value
Frequency band	915 MHz ISM (902–928 MHz)
Spreading factor	SF7
Bandwidth	125 kHz
Coding rate	4/5
Approximate data rate	5.47 kbps
Max payload per packet	245 bytes (payload)

E. Results

Table II summarizes transmission outcomes for a 10-second speech clip.

TABLE II
Transmission Results for a 10-Second Speech Clip at 48 kHz

Metric	Raw Audio	Compressed (EnCodec)	Reduction
File size	946 kB	4 kB	−99.6%
Fragment count	7,391	32	−99.6%
Transmission time	≈ 104 min	≈ 25 s	−99.6%
Intelligibility	Original	Intelligible (speaker ID preserved)	—

Semantic compression reduced file size by 99.6%, cutting packet count from 7,391 to 32 and transmission time from approximately 104 minutes to 25 seconds, bringing a 10-second voice message within practical airtime and duty-cycle limits of LoRa operation.

Reconstructed audio was subjectively intelligible across the full 10-second clip with speaker identity preserved. Observed artifacts included mild ringing and softened consonant onsets, consistent with low-bitrate neural coding at two RVQ codebook layers. These artifacts did not impair comprehension in subjective listening tests. No formal Mean Opinion Score (MOS) evaluation per ITU-T P.800 [13] was conducted in this phase; formal perceptual evaluation is planned as future work.

V. Conclusion

This paper presented a semantic voice compression system for peer-to-peer LoRa communication in infrastructure-less environments. By combining EnCodec [9] with a lightweight fragmentation and session protocol on commodity ESP32S3 hardware, the system achieves 99.6% compression, reducing a 10-second voice message from 7,391 packets to 32 and transmission time from over 100 minutes to 25 seconds, while preserving speech intelligibility and speaker identity. Unlike STT-to-TTS approaches, this method retains the prosody and urgency cues that matter most in emergency communication.

Several limitations bound the current validation. Tests were conducted indoors at short range; real-world link budgets, longer-distance operation at higher spreading factors, and formal perceptual evaluation per ITU-T P.800 [13] remain as future work. EnCodec inference currently requires GPU-class hardware, and the system is limited to a single-hop peer-to-peer link.

Future work will focus on migrating inference to on-device edge hardware, extending the transport to a multi-hop LoRa mesh [5], [6], and incorporating a dual-stream semantic encoder [10], [12] to further reduce payload while preserving prosodic content.

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